

RYAN HARTUNG

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EDUCATION

The University of Notre Dame

Ph.D. in Computer Science & Engineering

- Advisor: Dr. Douglas Thain
- Research focus: Distributed Systems and High Performance Computing

Notre Dame, IN

Expected: 2029

The University of Notre Dame

Master of Science in Computer Science & Engineering

Notre Dame, IN

August 2025

The University of Texas at Austin

Bachelor of Science in Physics; Computation Option

- Certificate in Elements of Computing

Austin, TX

May 2024

TEACHING

The University of Notre Dame

Graduate Teaching Assistant

- Elements of Computing II
- Principles of Computing

Notre Dame, IN

August 2024 - May 2025

Spring 2025

Fall 2024

PUBLICATIONS

1. xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture

- Authors: Liubov Kurafeeva, Alan Subedi, **Ryan Hartung**, Michael Fay, Avhishek Biswas, Shantenu Jha, Ozgur Kilic, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, Rich Wolski
- Link: <https://doi.org/10.1145/3731599.3767589>
- Published: SC Workshops '25: Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis Pages 2317 - 2327

TALKS/PANELS

(Nov. 2025) Invited panelist at **XLOOP Workshop at SC '25** [St. Louis, MO] ==> [Watch Here](#)

WORK EXPERIENCE

The University of Notre Dame

Graduate Research Assistant

- Assisting with the development of [xGFabric](#).

Notre Dame, IN

January 2025 - Present

Hargrove Engineers and Constructors

Controls & Automations Engineering Intern

- Automated the migration of 200,000+ lines of Dow MOD5 code into Honeywell Experion without the loss of functionality
- Tracked, analyzed, and documented over 7,000 variables to design and draw comprehensive flow diagrams to visualize variable definitions
- Developed multiple algorithms to expedite the migration and variable mapping processes, improving overall efficiency

Katy, TX

May 2024 - August 2024

Hargrove Engineers and Constructors

Controls & Automations Engineering Intern

- Instantiated existing library modules to complete the upcoming Reactor and Train configuration for Westlake's PVC plant

Katy, TX

May 2023 - August 2023

- Performed safety testing on all module and sequence configurations using internal procedures on simulated ABB hardware
- Successfully configured Modbus communication between the ABB and HIMA system controllers
- Updated structured text configuration to handle cross-controller communications

The University of Texas at Austin

Austin, Texas

Undergraduate Research Assistant

June 2022 - December 2023

- Created an open-source program which leverages various Python packages to analyze star-spots patterns across multiple thousands of stars in a compressed algorithm. Ensured that the algorithm was expandable and efficient for large datasets (over 6,000 star systems)
- Compared the contrast ratio of star-spots from two different NASA missions (Kepler and TESS) to establish a basis for exoplanet identification
- Implemented parallel computing and multi-core processing to optimize runtime and compilation
- Published results to a public [GitHub repository](#)
- Funded by a NASA research grant

LEADERSHIP AND INVOLVEMENT

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| • Society of Catholic Scientists | August 2025 - Present |
| • Science Policy Initiative at Notre Dame | August 2025 - Present |
| • Catholic Graduate Community | April 2025 - Present |
| • Technology and Catholicism Club | February 2025 - Present |
| • Notre Dame Rock Climbing Club Member | August 2024 - Present |
| • Notre Dame Texas Club Member | August 2024 - Present |
| • Science Olympiad (Competitor, Mentor, Club President, Proctor, Event-Supervisor, and Lab Coordinator) | August 2011 - Present |

TECHNICAL SKILLS

- Proficient Programming Languages: C/C++, HTML, Java, JavaScript, JSON, Julia, Python, Rust, XML
- Machine Learning & AI: Keras, PyTorch, Scikit-learn, TensorFlow
- Distributive Systems: Univa Grid Engine (UGE), HTCondor, SLURM
- Other Tools & Frameworks: ABB, Git, Honeywell, LaTeX, Linux/UNIX, Microsoft Suite, OpenFOAM, ParaView